

Muhammad Abban Fahim

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About me

I am a final year undergraduate student of robotics engineering, with research interests in machine vision and reinforcement learning and their applications to autonomous systems. With multiple years of software engineering experience, strong theoretical knowledge, and a constant drive to learn and try new things, I am an aspiring researcher and engineer, looking to apply robotics to real-life problems and products.

Education

Heriot-Watt University, BEng (Hons) - Robotics, Autonomous and Interactive Systems Sept 2022 – June 2026

- Current GPA: First-Class Honours (87/100)
- **Thesis:** Vision-based reinforcement learning for quadruped stair climbing

Skills

Languages: C++, C, Python, Javascript, C#, Rust, Lua, Java

Technologies

ROS	Unity	MATLAB	PyTorch	OpenCV	PlatformIO
Linux	Gazebo	Docker	Svelte	ReactJS	NodeJS
		CI/CD	OpenGL		

Experience

Undergraduate research assistant, Heriot-Watt University – Dubai, UAE

- Supervised by Dr. Claudio Zito at the Laboratory of AI and Robotics Research (August 2025 - Present):
 - Investigating probabilistic grasp synthesis by fusing object contact features with hand configuration priors.
 - Improving generalisation of grasps for unknown objects using partial point-clouds and few-shot demonstrations.
 - Developed ROS2 packages and visualization tools to capture training data and inspect inference likelihoods in real-time.
- Supervised by Dr. Nidhal Abdulaziz (July 2024 - May 2025):
 - Developed a surface-water cleaning USV, focusing on vision-aided autonomous navigation and localisation.
 - Administered the software engineering aspects of the USV, and developed a photorealistic hardware-in-the-loop simulation with Unity and ROS to enable testing before deployment.
 - Drafted design and budget proposals to secure funding for prototype.

Robotics software intern, Robotair – Remote

July 2025 – August 2025

- Contributed to open-source Python libraries (Rigel) for orchestrating and managing ROS applications using Docker.
- Tested and improved tooling for unit-testing, CI/CD pipelines, and monitoring robot deployments.
- Created end-user and internal documentation for codebases.
- Researched and presented implementation ideas for new product features for remote robot inspection.

President, E4L Society, Heriot-Watt University – Dubai, UAE

Sept 2024 – March 2025

- Organised the university's engineering society, from managing events, educational workshops and competitions for students to learn and demonstrate engineering skills, to leading these workshops.
- Recruited and led a team to manage the society's finances, planning, and outreach.

- Awarded Best Academic Society for 2024-25.

Simulation engineer intern, Alp Autonomy – Remote

June 2024 – Sept 2024

- Developed a hardware-in-the-loop simulator for UAVs, to test flight planning, and generate synthetic data for training deep-learning image segmentation models.
- Gained experience using the Unity game engine, OpenCV, PyTorch and ROS.

Publications

SUSHI: A Vision System for Reactive, Uninformed ASV Navigation via Multi-Field Path Planning and Visual Exploration in *Journal of Intelligent & Robotic Systems*

February 2026

Hamze Hammami, **Muhammad Abban**, Abdul Maajid Aga, Laith Mohamed, Saif Alsaad, Nidhal Abdulaziz

DOI: 10.21203/rs.3.rs-7490170/v1

Projects

BURGER trash collection robot

[LinkedIn link](#)

- Developed a surface-water cleaning USV, focusing on vision-aided autonomous navigation and localisation.
- Responsible for developing the simulation, used for prototyping the vision algorithms and integrating with other sensors.
- Worked on developing ROS nodes and micro-ROS code for interfacing with sensors and thruster control.
- Led and managed the team's software development through version control and Github organizational tools.

HAZMAT SLAM robot

[GitHub link](#)

- Designed and built a mobile robot with 3-DOF manipulator and gripper from scratch.
- Developed algorithms for color following, object detection, and visual servoing to enable autonomous operations for the robot.
- Implemented SLAM through Cartographer and the Nav2 stack to demonstrate navigation and obstacle-avoidance.

Tic-tac-toe robot

[GitHub link](#)

- Developed programs with ROS2 and Moveit2 to interface with a manipulator, and V-REP (CoppeliaSim) for simulating a test scene.
- Implemented low-level tasks from scratch with IK and FK.
- Integrated computer vision to allow multiple agents to play tic-tac-toe against the manipulator.

A2RL STEM Competition 2024

[LinkedIn link](#)

- Participated and won competition to develop autonomous racing software for miniature RC cars.
- Implemented theory of PID control, dynamic models, using Python and ROS, to optimise offline trajectories, and mathematical techniques to learn and improve control parameters.

CIE MCQ Website

[GitHub link](#)

- Developed a responsive web app for solving CAIE IGCSE and A-level MCQ questions for multiple subjects.
- Implemented real-time feedback, time-tracking and PDF processing.
- Improved results for over a 1000 students, and used feedback to improve and prioritise new app features.

Awards

A2RL Stem Competition, First place

- Winner for achieving the fastest track time and winning in all head-to-head races.

Curtin University Dubai Hackathon 2022, Second place

- Winner in one-day hackathon with a series of challenges involving Python and web development.